🎓 Final Examination – AI & Data Science (Project-Based)

🧠 Exam Overview

This examination is designed to assess your ability to apply **Data Science, Machine Learning, Deep Learning, and modern AI concepts** in a **real-world healthcare scenario**.

You will work as a **Data Scientist in a hospital network**, where your task is to analyze patient data, build predictive models, and propose AI-driven solutions to improve healthcare outcomes.

🏥 Industry Scenario

Hospitals often face a critical challenge:

> **Patients being readmitted within 30 days of discharge**

This leads to:

- Increased healthcare costs
- Poor patient outcomes
- Operational inefficiencies

Your role is to build an **AI-powered system** that can:

- Identify high-risk patients
- Predict readmissions
- Assist doctors in decision-making
- Enable intelligent automation

📊 Dataset

You will use the following real-world dataset:

👉 Diabetes 130-US Hospitals Dataset

Dataset Link:
https://www.kaggle.com/datasets/brandao/diabetes

📁 Dataset Description

This dataset contains patient records collected from multiple hospitals, including:

- Patient demographics
- Admission and discharge details
- Diagnoses and procedures
- Medications
- Lab results
- Length of stay
- **Readmission status (Target Variable)**

🎯 Examination Tasks

🔹 Part 1: Data Understanding & Preprocessing (15 Marks)

- Load the dataset using appropriate Data Science libraries
- Display:
  - First five records
  - Dataset shape and column names
- Handle missing or inconsistent values
- Perform necessary data cleaning and encoding

🔹 Part 2: Exploratory Data Analysis (EDA) (20 Marks)

Perform analysis and provide insights:

- Which patients are more likely to be readmitted?

- Does length of stay influence readmission?
- Identify key patterns or trends affecting readmission

Provide both **code and interpretation**.

🔹 Part 3: Scenario-Based Data Visualization (20 Marks)

Create visualizations and justify them based on the following real-world scenarios:

🟢 Scenario 1: Clinical Decision Support

Doctors want to identify **high-risk patients before discharge**.

- Select an appropriate visualization
- Create the chart
- Explain why it is suitable for clinical use

🟢 Scenario 2: Hospital Management

Management asks:

> “Which factors contribute most to patient readmission?”

- Choose suitable visualization(s)
- Present findings clearly
- Explain business/medical insights

🟢 Scenario 3: Executive Dashboard

Hospital executives require a **high-level summary of readmission trends**.

- Select simple and effective visualizations
- Justify your choice
- Explain how it supports decision-making

🔹 Part 4: Machine Learning (20 Marks)

Develop a predictive model to:

👉 **Predict Patient Readmission (Yes/No)**

Steps:

- Split dataset into training and testing sets
- Train a suitable classification model
- Evaluate performance using appropriate metrics
- Briefly interpret results

🔹 Part 5: Deep Learning (Practical Design) (10 Marks)

Answer the following:

- Why would Deep Learning be useful for large-scale hospital data?
- Design a simple Neural Network architecture:
  - Input layer (features used)
  - Hidden layers (conceptual)
  - Output layer

Provide a **clear structure or pseudo-implementation**.

🔹 Part 6: Generative AI (Applied Use Case) (10 Marks)

The hospital wants to improve patient communication.

- Write a **prompt** for a Generative AI system to:
  - Generate personalized discharge or follow-up instructions for a high-risk patient

Include:

- Sample patient details
- Prompt
- Expected output

🔹 Part 7: Agentic AI (Automation Design) (5 Marks)

Design an **AI Agent workflow** for hospital automation:

- Define:
  - Input
  - Process
  - Output

Example use case:

- Detect high-risk patient → Notify doctor → Schedule follow-up → Send reminder

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🔹 Part 8: Bonus (Optional – 5 Marks)

Propose one AI-based solution that can:

- Reduce hospital readmissions
  OR
- Improve patient care efficiency

📁 Submission Requirements

- Python Notebook (.ipynb) or Python script (.py)
- All outputs and visualizations must be included
- Explanations should be clearly written within the notebook

🧪 Evaluation Criteria

Component	Marks
Data Preprocessing	15
EDA & Insights	20

Visualization (Scenario-Based)	20
Machine Learning Model	20
Deep Learning Design	10
Generative AI Application	10
Agentic AI Design	5
Total	**100 Marks**